Digital VLSI Design Assignment: 4-bit Array Multiplier from RTL to GDSII with Open-Source Tools

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1 Introduction

A 4-bit unsigned array multiplier (16 AND gates, 12 full adders, 16 flip-flops, latency 2 clock cycles) was written in Verilog, verified exhaustively (256 vectors), synthesised with Yosys, timed with OpenSTA and taken through LibreLane to a GDSII on SkyWater SKY130 (sky130_fd_sc_hd). All tools ran in the IIC-OSIC-TOOLS Docker image. Every number in this report was generated from the files in the repository by report/tools/make_report.py. AI-assistant disclosure: AI assistants (ChatGPT and Claude) were used for guidance on the tool flow, scripting and for drafting; the code was run, checked and understood by us.

2 Design and Verification (Part A)

The design registers a and b (8 flip-flops), builds the array from the registered values (row 0 only ANDs; rows 1-3 each 4 AND + 4 full adders in a ripple chain) and registers the 8-bit product (8 flip-flops). Reset is asynchronous and active low.

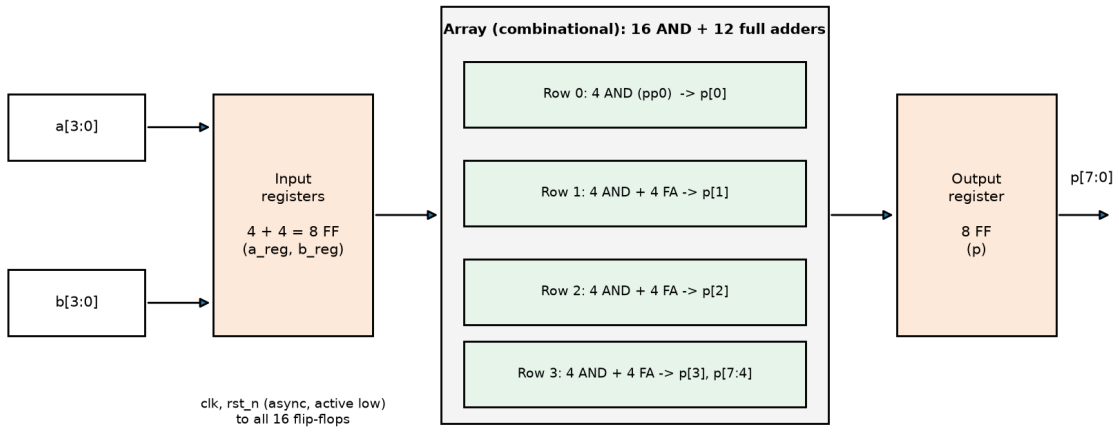


Figure 1: Block diagram: 16 AND gates, 12 full adders and the 16 flip-flops.

Simulation results

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Correct design : PASS: 256 vectors checked, 0 errors
Fault test     : FAIL: 256 vectors checked, 95 errors   (full_adder.v with sum = a ^ b)
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Example (roll number 2026203012): A = last digit + 5 = 7, B = 9; hand-worked in the layout of Section 2:

```
Roll number 2026203012: last digit 2  ->  A = 2 + 5 = 7,  B = 9  (A = 7, so B = 9)

          . . . . 0 1 1 1   A = 7
x         . . . . 1 0 0 1   B = 9
          ---------------
          . . . . 0 1 1 1   pp0 = A and b0 (b0 = 1), shifted 0
          . . . 0 0 0 0 .   pp1 = A and b1 (b1 = 0), shifted 1
          . . 0 0 0 0 . .   pp2 = A and b2 (b2 = 0), shifted 2
          . 0 1 1 1 . . .   pp3 = A and b3 (b3 = 1), shifted 3
          ---------------
          0 0 1 1 1 1 1 1   P = 63 = 0x3F
```

Figure 2: 7 x 9 = 63 (0x3F) worked out with partial products.

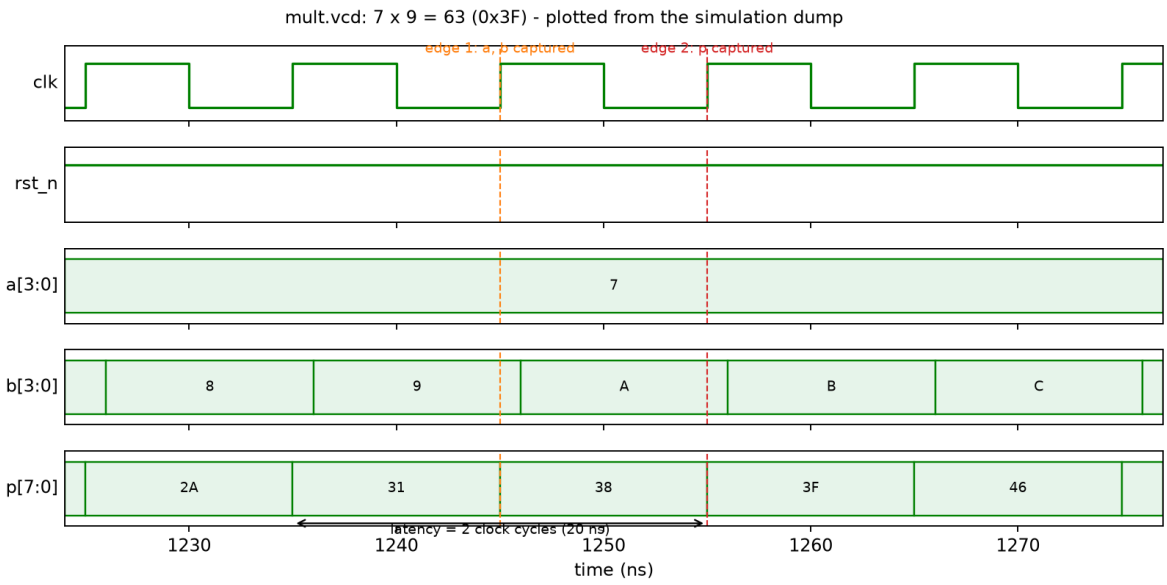


Figure 3: Waveform of 7 x 9: p = 0x3F appears two clock edges after a, b are applied (latency 2 cycles). Plotted from mult.vcd.

3 Synthesis (Part B)

synth/synth.tcl performs the seven steps of the assignment (read_verilog, synth -top mult_array -flatten, dfflibmap, abc, opt_clean -purge, stat, write_verilog).

Metric	Value
Total number of cells	63
Number of flip-flops	16 (dfrtp_1)
Total cell area (um2)	740.710
Flip-flop area (um2)	400.384 (16 x 25.024)
Flip-flop area (% of total)	54.05 %
3 most-used cell types	dfrtp_1: 16, nand2_1: 8, a21oi_1: 6
Netlist simulation result	see below (PASS line)

Table 1: Part B results (from synth/mult_array_stat.rpt).

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Gate-level simulation, assignment command (with -DUNIT_DELAY=#1): FAIL: 256 vectors checked, 239 errors
Gate-level simulation, without -DUNIT_DELAY           : PASS: 256 vectors checked, 0 errors
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-DUNIT_DELAY=#1 gives every gate a 1 ns delay; if the result of that run is FAIL it is only because the artificial delays make the combinational path longer than the 10 ns testbench clock, not because synthesis changed the function. Without the artificial delay the netlist is checked at zero gate delay.

4 Timing (Part C)

sta/sta.tcl reads the typical-corner library (tt_025C_1v80), links the netlist, defines clk with period T, sets 1.0 ns input and output delays and reports the worst slack and the critical path. At T = 10 ns the worst slack is 7.396 ns, so Tmin = 10 - 7.396 = 2.604 -> 2.60 ns (rounded up to 0.01 ns) and Fmax = 1000/2.60 = 384.6 MHz.
Re-running at T = 2.60 ns gives a slack of 0.016 ns.

Metric	Value
Worst slack at T = 10 ns (ns)	7.396
Tmin (ns)	2.60
Fmax (MHz)	384.6

Metric	Value
Slack at T = Tmin (ns)	0.016
Critical path start point	_095_ = a_reg[1]
Critical path end point	_092_ = p[6]
Number of cells on critical path	8 logic cells (10 with the two flip-flops)
Latency (ns) = 2 x Tmin	5.20
Throughput (million results/s)	384.6

Table 2: Part C results.

#	Cell	Delay (ns)	Arrival (ns)
launch	dfrtp_1 (_095_), clk->Q	0.398	0.398
1	clkinv_1 (_039_)	0.074	0.472
2	o311ai_0 (_047_)	0.348	0.820
3	a21oi_1 (_050_)	0.172	0.992
4	maj3_1 (_063_)	0.381	1.373
5	xnor2_1 (_064_)	0.160	1.534
6	maj3_1 (_066_)	0.385	1.918
7	maj3_1 (_075_)	0.377	2.295
8	xnor3_1 (_079_)	0.175	2.471
capture	dfrtp_1 (_092_), setup 0.134		slack 0.016

Table 3: Critical path a_reg[1] -> p[6] at T = 2.60 ns. 5 of the 8 logic cells are adder-type cells (xor/xnor/maj).

5 Physical Design (Part D)

pnr/config.json sets DESIGN_NAME mult_array, the two Verilog files, CLOCK_PORT clk, PDK sky130A, STD_CELL_LIBRARY sky130_fd_sc_hd, FP_SIZING absolute and DIE_AREA [0, 0, 150, 150], plus SYNTH_STRATEGY DELAY 2. The starting period was 1.2 x Tmin rounded up to the next 0.5 ns, then raised until setup slack was non-negative in every corner LibreLane checks. The final period is 7.5 ns.

Metric	Value
CLOCK_PERIOD used (ns)	7.5
Number of runs needed	22 LibreLane runs in total (all listed in Table 5); the first passing period was 7.5 ns
Core area (um2)	17759.5
Cells after synthesis (Part B)	63
Cells after place and route	1961 instances in total = 386 standard cells (incl. 245 tap) + 1575 fill; 141 non-physical cells (logic, flip-flops, buffers)
Kinds of cells added	tap 245, fill 1575, timing-repair/IO buffers 27, clock buffers 3, decap 0
Wirelength (um)	3778 (routed), 967 vias
Setup slack (ns)	0.173 (worst corner ss_100C_1v60); 3.714 typical corner
Post-route Fmax (MHz)	136.5 (worst corner); 264.1 (typical corner)
Fmax from Part C (MHz)	384.6
DRC errors	Magic 0, KLayout 0, router 0; antenna violations 0
LVS errors	0

Metric	Value
Hold slack (ns)	0.137, 0 violations
Max slew / cap / fanout violations	4 / 0 / 0 (slew: only in the extreme slow corner, see note)

Table 4: Part D results (final run, from pnr/metrics.json).

Note on max-slew: the flow reports 4 maximum-slew violations, all in the max_ss_100C_1v60 corner (100 C, 1.60 V, slow transistors, worst-case RC). They are an additional electrical sign-off warning from the library's max_transition limit, not one of the assignment's pass criteria (DRC = 0, LVS = 0, setup slack >= 0). The typical corner has 0 slew, 0 cap and 0 fanout violations. We tried SYNTH_STRATEGY variants, SYNTH_SIZING, MFS3 and post-global-route design repair (see Table 5); because they changed timing or area without a clean result and relaxing limits only hides the warning, the submitted run keeps the plain flow.

Run	T (ns)	Synthesis strategy	Other settings	Setup WNS (ns)	Setup viol.	Hold WNS	Slew viol.	DRC	LVS	Result
runs/RUN_2026-10-24_14-01-01	3.5	AREA 0	SYNTH_STRATEGY=AREA 0	-2.132	15	0.119	0	0	0	setup/other FAIL
runs/delay2_test	3.5	DELAY 2	-	-1.666	15	0.095	6	0	0	setup/other FAIL
runs/delay4_test	3.5	DELAY 4	SYNTH_STRATEGY=DELAY 4	-2.714	15	0.107	11	0	0	setup/other FAIL
runs/RUN_2026-10-24_20-05-46	3.5	DELAY 2	PL_RESIZER_SETUP_MAX_BUFFER_PCT=100; GRT_RESIZER_SETUP_MAX_BUFFER_PCT=100	-1.666	15	0.095	6	0	0	setup/other FAIL
runs/delay2_util80_test	3.5	DELAY 2	PL_RESIZER_SETUP_MAX_UTIL_PCT=80; GRT_RESIZER_SETUP_MAX_UTIL_PCT=80	-1.666	15	0.095	6	0	0	setup/other FAIL
runs/delay2_keepbuf_test	3.5	DELAY 2	PL_RESIZER_SETUP_BUFFER_REMOVAL=False; GRT_RESIZER_SETUP_BUFFER_REMOVAL=False	-1.666	15	0.095	6	0	0	setup/other FAIL
runs/RUN_2026-10-24_21-15-52	3.5	DELAY 2	SYNTH_SIZING=True	-1.666	15	0.095	6	0	0	setup/other FAIL
mfs3_test/runs/RUN_2026-10-24_21-47-34	3.5	DELAY 2	SYNTH_ABC_USE_MFS3=True	-1.777	15	0.065	11	0	0	setup/other FAIL
mfs3_test/runs/RUN_2026-10-24_21-50-41	3.5	DELAY 2	SYNTH_ABC_USE_MFS3=True	-1.777	15	0.065	11	0	0	setup/other FAIL
runs/RUN_2026-10-25_05-25-21	5.8	DELAY 2	-	-0.587	9	0.135	11	0	0	setup/other FAIL
runs/RUN_2026-10-25_05-32-46	7.5	DELAY 2	-	0.144	0	0.137	15	0	0	PASS
runs/RUN_2026-10-25_05-59-40	7.5	DELAY 2	RUN_POST_GRT_DESIGN_REPAIR=True; GRT_DESIGN_REPAIR_MAX_SLEW_PCT=20	0.173	0	0.137	4	0	0	PASS
runs/RUN_2026-10-25_06-06-56	7.5	DELAY 2	RUN_POST_GRT_DESIGN_REPAIR=True; GRT_DESIGN_REPAIR_MAX_SLEW_PCT=20	0.173	0	0.137	4	0	0	PASS

Run	T (ns)	Synthesis strategy	Other settings	Setup WNS (ns)	Setup viol.	Hold WNS	Slew viol.	DRC	LVS	Result
runs/RUN_2026-10-25_06-13-56	4	DELAY 2	-	-1.166	15	0.134	6	0	0	setup/other FAIL
runs/RUN_2026-10-25_06-30-48	4	DELAY 2	-	-1.166	15	0.134	6	0	0	setup/other FAIL
runs/RUN_2026-10-25_06-36-14	5	DELAY 2	-	-0.353	9	0.136	6	0	0	setup/other FAIL
runs/RUN_2026-10-25_06-41-37	5.5	DELAY 2	-	-0.484	9	0.137	6	0	0	setup/other FAIL
runs/RUN_2026-10-25_06-47-13	6	DELAY 2	-	-0.624	9	0.137	15	0	0	setup/other FAIL
runs/RUN_2026-10-25_07-03-11	7	DELAY 2	-	-0.356	7	0.137	15	0	0	setup/other FAIL
runs/partD_3p5ns	3.5	DELAY 2	-	-1.666	15	0.095	6	0	0	setup/other FAIL
runs/partD_3p5ns_repair	3.5	DELAY 2	RUN_POST_GRT_DESIGN_REPAIR=True; GRT_DESIGN_REPAIR_MAX_SLEW_PCT=20	-1.639	15	-0.354	0	0	0	setup/other FAIL
runs/partD_3p5ns_delay3	3.5	DELAY 3	RUN_POST_GRT_DESIGN_REPAIR=True; SYNTH_STRATEGY=DELAY 3; GRT_DESIGN_REPAIR_MAX_SLEW_PCT=20	-1.994	15	-0.193	0	0	0	setup/other FAIL

Table 5: Every LibreLane run found in pnr/ (from each run's final/metrics.json and resolved.json).

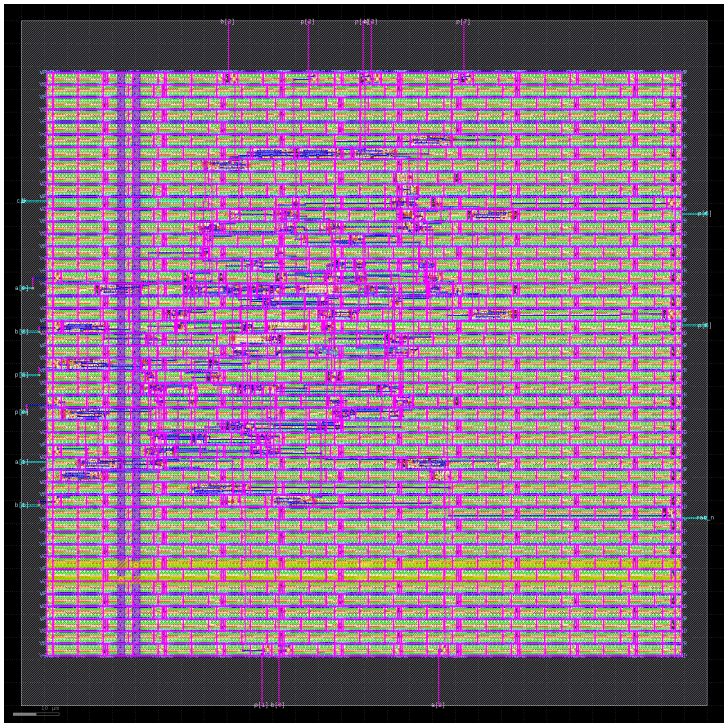


Figure 4: Whole die, 150 um x 150 um (KLayout).

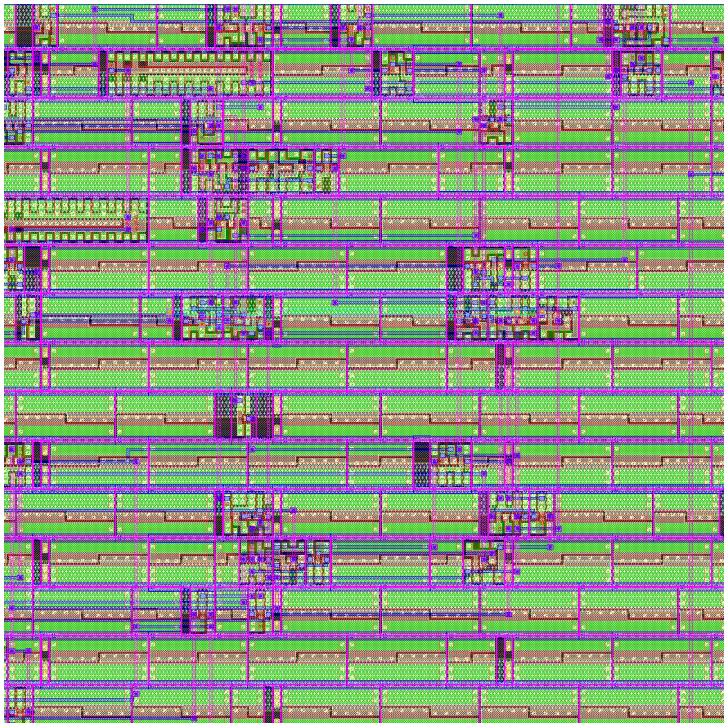


Figure 5: Zoomed view (40 um x 40 um): standard-cell rows and metal wires.

6 Answers to the Questions

1. Synthesis produced 63 cells in total, of which 16 are flip-flops, so the logic uses 47 cells, not 16 + 12 = 28. The 28 counts RTL operators, but Yosys flattens the design and ABC re-maps the whole logic cloud onto library gates. One full adder is not one gate: it needs a sum function (an XOR/XNOR of three inputs) and a carry function (a majority function), and the library offers them as separate cells or as small gate networks. The partial-product ANDs are merged into neighbouring adder logic (NAND/AOI/OAI style cells), the first adder of every row has carry-in 0 and collapses to a half adder, and extra inverters are added where a cell produces the opposite polarity. The mapping is therefore driven by area/delay, not by the RTL gate count.

2. The critical path (clk group, T = 2.60 ns) runs from register a_reg[1] (_095_) to register p[6] (_092_). It contains 8 logic cells (10 including the launch and capture flip-flops) with 2.072 ns of logic delay; adding clock-to-Q 0.398 ns and setup 0.134 ns gives 2.604 ns, which matches Tmin = 2.60 ns. The textbook estimate is about 2N = 8 full-adder delays. In the mapped netlist only 5 of the 8 cells on the path are adder-type cells (xor/xnor/maj/fa), the others are partial-product and merged logic, and the average delay is 0.259 ns per cell, because ABC restructures the ripple chain and does not keep eight separate full-adder blocks.

3. The 16 flip-flops occupy 400.384 um2 of the 740.710 um2 total, i.e. 54.1 %. One dfrtp_1 is 25.024 um2, about 6.7x the area of a nand2_1 (3.754 um2). The multiplier registers all 8 input bits and all 8 output bits, but its logic is only 47 small cells (340.3 um2), so the large asynchronous-reset flip-flops dominate even in a tiny design.

4. Post-route Fmax = 1000/(T - slack) = 1000/(7.5 - 0.173) = 136.5 MHz at the worst (slow, ss_100C_1v60) corner and 264.1 MHz at the typical corner, against 384.6 MHz from Part C. Part C used ideal clocks, zero wire delay, only the typical corner and the unplaced netlist; after routing there is wire RC (3778 um of routed wire), a real clock tree, repair buffers, a slow 100 C / 1.60 V corner, and the flow only has to meet the relaxed 7.5 ns target, so the slack-based Fmax is a conservative figure. The flow added 245 tap cells (tie wells to VDD/VSS and prevent latch-up), 1575 fill cells (make the rows, wells and power rails continuous and satisfy density rules), 27 timing-repair/IO buffers (fix slew, load and timing) and 3 clock buffers (distribute clk to all 16 flip-flops with low skew); no separate decap cells appear in the metrics.

5. DRC (design-rule check) verifies the layout geometry against the foundry's manufacturing rules: minimum width and spacing, enclosure, density and antenna rules. LVS (layout versus schematic) extracts the transistors and connections from the layout and compares them with the netlist, proving that the drawn chip is the circuit that was designed. A DRC failure means the layout may not be manufacturable or may have poor yield; an LVS

failure means a manufacturable chip could still be the wrong circuit. Both must be clean before tape-out. This run has 0 Magic DRC errors, 0 KLayout DRC errors and 0 LVS errors.

6. For N = 8 the array needs N^2 = 64 AND gates (4x the 16 of N = 4) and N(N-1) = 56 full adders (4.7x the 12). The critical path is about 2N full-adder delays, i.e. 16 instead of 8, so the combinational delay roughly doubles while area grows roughly quadratically. Scaling the measured logic delay: logic = 2 x 2.072 = 4.144 ns, so Tmin(8-bit) is about 0.398 + 4.144 + 0.134 = 4.68 ns, Fmax about 214 MHz (typical corner, same per-stage delay).

7 Conclusion

The multiplier passes all 256 vectors in RTL and as a netlist (the fault test is detected), reaches Tmin = 2.60 ns (Fmax = 385 MHz, typical corner) before layout, and produces a DRC/LVS-clean layout at 7.5 ns with 0.173 ns worst setup slack over all corners. Registers take 54.1 % of the cell area; the physical flow adds mainly tap and fill cells.